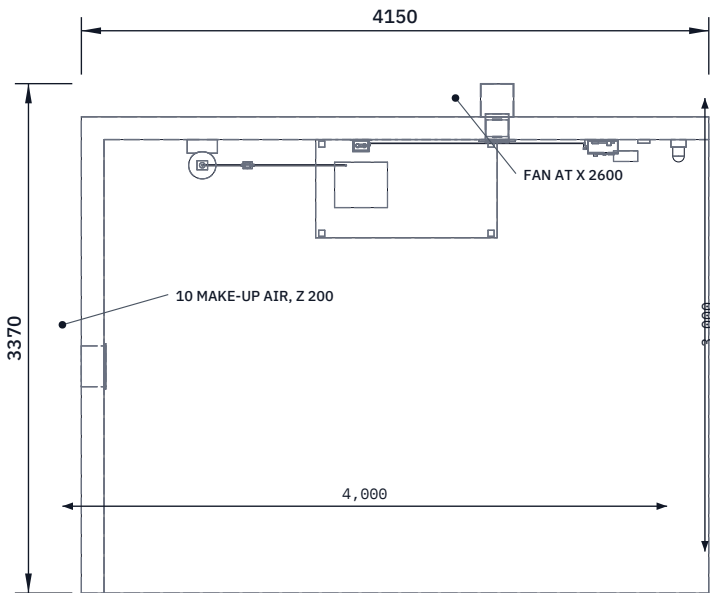


CONCEPT, NOT FOR FABRICATION

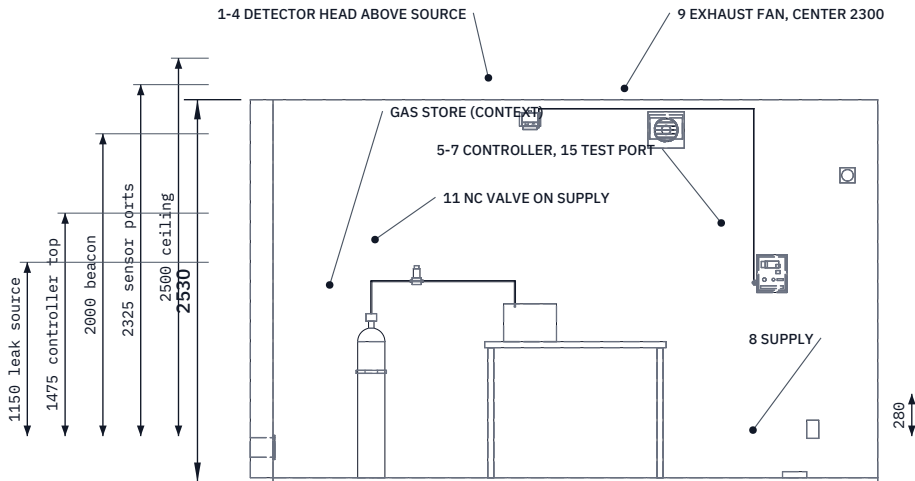
H2Guard · github.com/BoujeeEnjinia1701/h2guard

REV	DESCRIPTION	DATE	BY
P1	Preliminary GA for TRL 3 (from cad/src/model.py)	2026-09-25	AC
P2	Notes: head placement rule, timed escalation (DDR-002)	2026-09-25	AC



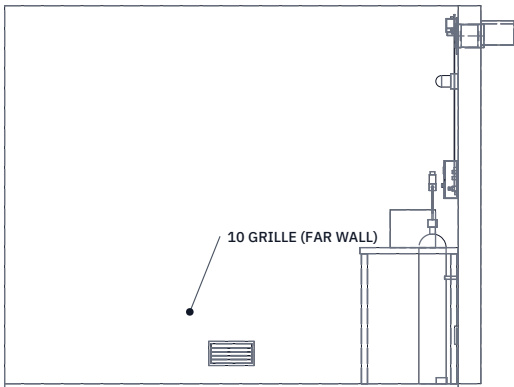
TOP VIEW

Scale 1:50; looking down (along -Z)



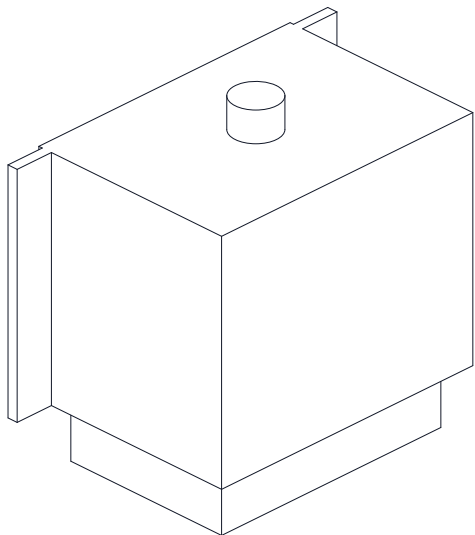
FRONT VIEW

Scale 1:50; looking at the front (along +Y)



RIGHT VIEW

Scale 1:50; looking at the right side (along -X)



DETECTOR HEAD, ITEMS 1 TO 4

Not to scale; ports and bump test cup face down

MAIN DIMENSIONS AND INTERFACES (MM)

Room 4000 x 3000 x 2500 (30 m3); leak source 1150 above floor
Head 110 x 80 x 90; ports 175 below ceiling, 1175 above source
Arrestor discs 25 x 2; sensor 2.2 behind disc
Bump cup 96 x 66 x 22 (114 mL); tube 4 OD, 2,695 run
Controller 200 x 250 x 90; top 1,025 below ceiling
Fan 150 duct, sleeve 190; grille top 80 below ceiling
Make-up grille 300 x 160 on far wall, center 200
Valve 1/4 in NC after regulator; gas fitting by a competent person
Head directly above each likely leak point; offset limit set at TRL 4
Warning held 5 min closes valve and latches (firmware, DDR-002)
Third-angle; front view from -Y (room side); HGD-CAL-001

H2GUARD

github.com/BoujeeEnjinia1701/h2guard

General arrangement in the 30 m3 reference room

DRAWING NO. HGD-DWG-001			REV P2	SHEET 1 of 1	SIZE ANSI B
SCALE 1:50	UNITS mm	PROJECTION 	DATE 2026-09-25		DRAWN Amish Chadha
MATERIAL / NOTES Bought-in parts per bom/bom.csv; room, bench and gas store are context. PRELIMINARY, NOT FOR FABRICATION					
Licensed CERN-OHL-S-2.0 · Design Molecule Lab · designmolecule.com					Generated from cad/src; do not edit by hand